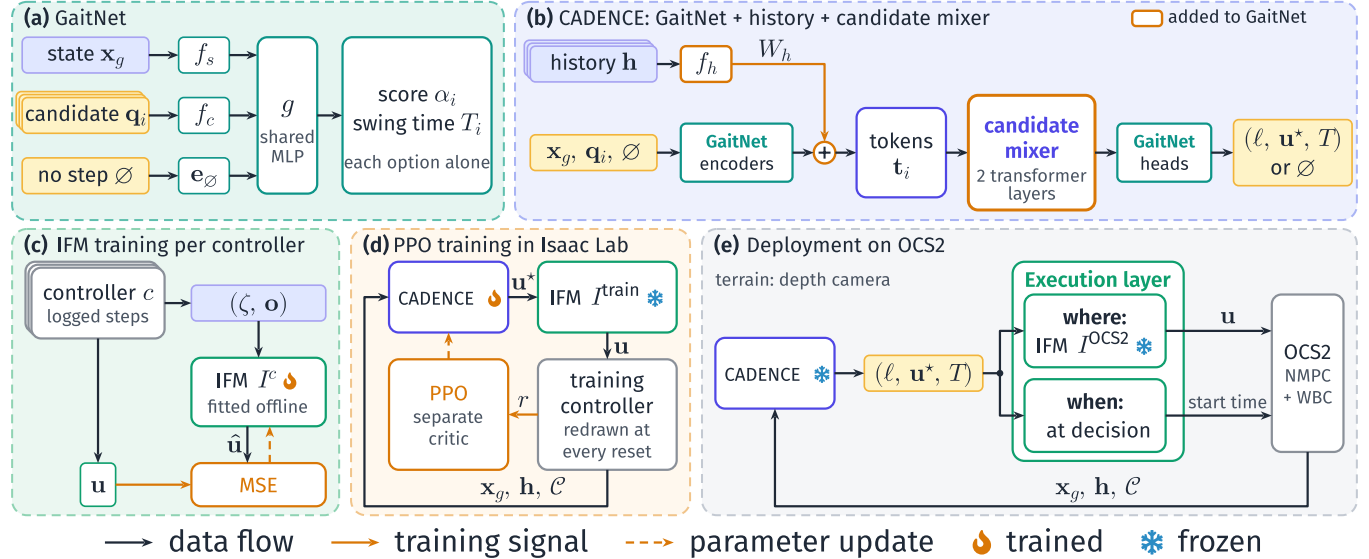

(a) GaitNet
state $\mathbf{x}_g$
candidate $\mathbf{q}_i$
no step $\varnothing$
f_s
f_c
$\mathbf{e}_\varnothing$
g
shared MLP
score α_i
swing time T_i
each option alone
(b) CADENCE: GaitNet + history + candidate mixer
added to GaitNet
history $\mathbf{h}$
f_h
W_h
$\mathbf{x}_g$, $\mathbf{q}_i$, $\varnothing$
GaitNet encoders
tokens $\mathbf{t}_i$
candidate mixer
2 transformer layers
GaitNet heads
$(\ell, \mathbf{u}^\star, T)$ or $\varnothing$
(c) IFM training per controller
controller c
logged steps
$(\zeta, \mathbf{o})$
IFM I^c
fitted offline
$\hat{\mathbf{u}}$
$\mathbf{u}$
MSE
(d) PPO training in Isaac Lab
CADENCE
$\mathbf{u}^\star$
IFM I^{train}
$\mathbf{u}$
PPO
separate critic
r
training controller
redrawn at every reset
$\mathbf{x}_g$, $\mathbf{h}$, $\mathcal{C}$
(e) Deployment on OCS2
terrain: depth camera
CADENCE
$(\ell, \mathbf{u}^\star, T)$
Execution layer
where: IFM I^{OCS2}
when: at decision
$\mathbf{u}$
start time
OCS2
NMPC + WBC
$\mathbf{x}_g$, $\mathbf{h}$, $\mathcal{C}$
data flow
training signal
parameter update
trained
frozen